

# A Unified Bias-Variance Decomposition

**Pedro Domingos**

Department of Computer Science and Engineering

University of Washington

Box 352350

Seattle, WA 98185-2350, U.S.A.

pedrod@cs.washington.edu

Tel.: 206-543-4229 / Fax: 206-543-2969

## **Abstract**

The bias-variance decomposition is a very useful and widely-used tool for understanding machine-learning algorithms. It was originally developed for squared loss. In recent years, several authors have proposed decompositions for zero-one loss, but each has significant shortcomings. In particular, all of these decompositions have only an intuitive relationship to the original squared-loss one. In this article, we define bias and variance for an arbitrary loss function, and show that the resulting decomposition specializes to the standard one for the squared-loss case, and to a close relative of Kong and Dietterich’s (1995) one for the zero-one case. The same decomposition also applies to variable misclassification costs. We show a number of interesting consequences of the unified definition. For example, Schapire et al.’s (1997) notion of “margin” can be expressed as a function of the zero-one bias and variance, making it possible to formally relate a classifier ensemble’s generalization error to the base learner’s bias and variance on training examples. We have applied the proposed decomposition to decision tree learning, nearest-neighbor learning and boosting on a large suite of benchmark datasets, and made several significant observations.

**Keywords:** Loss functions, bias, variance, decision trees, nearest neighbor, boosting

**Running head:** A Unified Bias-Variance Decomposition

# 1 Introduction

For the better part of the last two decades, machine-learning research has concentrated mainly on creating ever more flexible learners using ever more powerful representations. At the same time, very simple learners were often found to perform very well in experiments, sometimes better than more sophisticated ones (e.g., Holte (1993), Domingos & Pazzani (1997)). In recent years the reason for this has become clear: predictive error has two components, and while more powerful learners reduce one (bias) they increase the other (variance). The optimal point in this trade-off varies from application to application. In a parallel development, researchers have found that learning ensembles of models very often outperforms learning a single model (e.g., Bauer & Kohavi (1999)). That complex ensembles would outperform simple single models contradicted many existing intuitions about the relationship between simplicity and accuracy. This finding, apparently at odds with the one above about the value of simple learners, also becomes easier to understand in light of a bias-variance decomposition of error: while allowing a more intensive search for a single model is liable to increase variance, averaging multiple models will often (though not always) reduce it. As a result of these developments, the bias-variance decomposition of error has become a cornerstone of our understanding of inductive learning.

Although machine-learning research has been mainly concerned with classification problems, using zero-one loss as the main evaluation criterion, the bias-variance insight was borrowed from the field of regression, where squared-loss is the main criterion. As a result, several authors have proposed bias-variance decompositions related to zero-one loss (Kong & Dietterich, 1995; Breiman, 1996b; Kohavi & Wolpert, 1996; Tibshirani, 1996; Friedman, 1997). However, each of these decompositions has significant shortcomings. In particular, none has a clear relationship to the original decomposition for squared loss. One source of difficulty has been that the decomposition for squared-loss is purely additive (i.e.,  $\text{loss} = \text{bias} + \text{variance}$ ), but it has proved difficult to obtain the same result for zero-one loss using definitions of bias and variance that have all the intuitively necessary properties. Here we take the position that instead of forcing the bias-variance decomposition to be purely additive, and defining bias and variance so as to make this happen, it is preferable to start with a single consistent definition of bias and variance for all loss functions, and then investigate how loss varies as a function of bias and variance in each case. This should lead to more insight and to a clearer picture than a collection of unrelated decompositions. It should also make it easier to extend the bias-variance decomposition to further loss functions. Intuitively, since a bias-variance trade-off exists in any generalization problem, it should be possible and useful

to apply a bias-variance analysis to any “reasonable” loss function. We believe the unified decomposition we propose here is a step towards this goal.

We begin by proposing unified definitions of bias and variance, and showing how squared-loss, zero-one loss and variable misclassification costs can be decomposed according to them. This is followed by the derivation of a number of properties of the new decomposition, in particular relating it to previous results. We then describe experiments with the new decomposition and discuss related work and directions for future research.

## 2 A Unified Decomposition

Given a training set  $\{(x_1, t_1), \dots, (x_n, t_n)\}$ , a learner produces a model  $f$ . Given a test example  $x$ , this model produces a prediction  $y = f(x)$ . (For the sake of simplicity, the fact that  $y$  is a function of  $x$  will remain implicit throughout this article.) Let  $t$  be the true value of the predicted variable for the test example  $x$ . A *loss function*  $L(t, y)$  measures the cost of predicting  $y$  when the true value is  $t$ . Commonly used loss functions are squared loss ( $L(t, y) = (t - y)^2$ ), absolute loss ( $L(t, y) = |t - y|$ ), and zero-one loss ( $L(t, y) = 0$  if  $y = t$ ,  $L(t, y) = 1$  otherwise). The goal of learning can be stated as producing a model with the smallest possible loss; i.e., a model that minimizes the average  $L(t, y)$  over all examples, with each example weighted by its probability. In general,  $t$  will be a nondeterministic function of  $x$  (i.e., if  $x$  is sampled repeatedly, different values of  $t$  will be seen). The optimal prediction  $y_*$  for an example  $x$  is the prediction that minimizes  $E_t[L(t, y_*)]$ , where the subscript  $t$  denotes that the expectation is taken with respect to all possible values of  $t$ , weighted by their probabilities given  $x$ . The optimal model is the model for which  $f(x) = y_*$  for every  $x$ . In general, this model will have non-zero loss. In the case of zero-one loss, the optimal model is called the *Bayes classifier*, and its loss is called the *Bayes rate*.

Since the same learner will in general produce different models for different training sets,  $L(t, y)$  will be a function of the training set. This dependency can be removed by averaging over training sets. In particular, since the training set size is an important parameter of a learning problem, we will often want to average over all training sets of a given size. Let  $D$  be a set of training sets. Then the quantity of interest is the expected loss  $E_{D,t}[L(t, y)]$ , where the expectation is taken with respect to  $t$  and the training sets in  $D$  (i.e., with respect to  $t$  and the predictions  $y = f(x)$  produced for example  $x$  by applying the learner to each training set in  $D$ ). Bias-variance decompositions decompose the expected loss into three terms: bias, variance and noise. A standard such decomposition exists for squared loss, and

a number of different ones have been proposed for zero-one loss.

In order to define bias and variance for an arbitrary loss function we first need to define the notion of main prediction.

**Definition 1** *The main prediction for a loss function  $L$  and set of training sets  $D$  is  $y_m^{L,D} = \operatorname{argmin}_{y'} E_D[L(y, y')]$ .*

When there is no danger of ambiguity, we will represent  $y_m^{L,D}$  simply as  $y_m$ . The expectation is taken with respect to the training sets in  $D$ , i.e., with respect to the predictions  $y$  produced by learning on the training sets in  $D$ . Let  $Y$  be the multiset of these predictions. (A specific prediction  $y$  will appear more than once in  $Y$  if it is produced by more than one training set.) In words, the main prediction is the value  $y'$  whose average loss relative to all the predictions in  $Y$  is minimum (i.e., it is the prediction that “differs least” from all the predictions in  $Y$  according to  $L$ ). The main prediction under squared loss is the mean of the predictions; under absolute loss it is the median; and under zero-one loss it is the mode (i.e., the most frequent prediction). For example, if there are  $k$  possible training sets of a given size, we learn a classifier on each,  $0.6k$  of these classifiers predict class 1, and  $0.4k$  predict 0, then the main prediction under zero-one loss is class 1. The main prediction is not necessarily a member of  $Y$ ; for example, if  $Y = \{1, 1, 2, 2\}$  the main prediction under squared loss is 1.5.

We can now define bias and variance as follows.

**Definition 2** *The bias of a learner on an example  $x$  is  $B(x) = L(y_*, y_m)$ .*

In words, the bias is the loss incurred by the main prediction relative to the optimal prediction.

**Definition 3** *The variance of a learner on an example  $x$  is  $V(x) = E_D[L(y_m, y)]$ .*

In words, the variance is the average loss incurred by predictions relative to the main prediction. Bias and variance may be averaged over all examples, in which case we will refer to them as *average bias*  $E_x[B(x)]$  and *average variance*  $E_x[V(x)]$ .

It is also convenient to define noise as follows.

**Definition 4** *The noise of an example  $x$  is  $N(x) = E_t[L(t, y_*)]$ .*

In other words, noise is the unavoidable component of the loss, incurred independently of the learning algorithm.

Definitions 2 and 3 have the intuitive properties associated with bias and variance measures.  $y_m$  is a measure of the “central tendency” of a learner. (What “central” means depends on the loss function.) Thus  $B(x)$  measures the systematic loss incurred by a learner, and  $V(x)$  measures the loss incurred by its fluctuations around the central tendency in response to different training sets. The bias is independent of the training set, and is zero for a learner that always makes the optimal prediction. The variance is independent of the true value of the predicted variable, and is zero for a learner that always makes the same prediction regardless of the training set. However, it is not necessarily the case that the expected loss  $E_{D,t}[L(t, y)]$  for a given loss function  $L$  can be decomposed into bias and variance as defined above. Our approach will be to propose a decomposition and then show that it applies to each of several different loss functions. We will also exhibit some loss functions to which it does not apply. (However, even in such cases it may still be worthwhile to investigate how the expected loss can be expressed as a function of  $B(x)$  and  $V(x)$ .)

Consider an example  $x$  for which the true prediction is  $t$ , and consider a learner that predicts  $y$  given a training set in  $D$ . Then, for certain loss functions  $L$ , the following decomposition of  $E_{D,t}[L(t, y)]$  holds:

$$\begin{aligned} E_{D,t}[L(t, y)] &= c_1 E_t[L(t, y_*)] + L(y_*, y_m) + c_2 E_D[L(y_m, y)] \\ &= c_1 N(x) + B(x) + c_2 V(x) \end{aligned} \tag{1}$$

$c_1$  and  $c_2$  are multiplicative factors that will take on different values for different loss functions. We begin by showing that this decomposition reduces to the standard one for squared loss.

**Theorem 1** *Equation 1 is valid for squared loss, with  $c_1 = c_2 = 1$ .*

*Proof.* Substituting  $L(a, b) = (a - b)^2$ ,  $y_* = E_t[t]$ ,  $y_m = E_D[y]$  and  $c_1 = c_2 = 1$ , Equation 1 becomes:

$$E_{D,t}[(t - y)^2] = E_t[(t - E_t[t])^2] + (E_t[t] - E_D[y])^2 + E_D[(E_D[y] - y)^2] \tag{2}$$

This is the standard decomposition for squared loss, as derived in (for example) Geman et al. (1992).  $y_* = E_t[t]$  because  $E_t[(t - y)^2] = E_t[(t - E_t[t])^2] + (E_t[t] - y)^2$  (also shown in Geman et al. (1992), etc.), and therefore  $E_t[(t - y)^2]$  is minimized by making  $y = E_t[t]$ .  $\square$

Some authors (e.g., Kohavi and Wolpert, 1996) refer to the  $(E_t[t] - E_D[y])^2$  term as “bias squared.” Here we follow the same convention as Geman et al. (1992) and others, and

simply refer to it as “bias.” This makes more sense given our goal of a unified bias-variance decomposition, since the square in  $(E_t[t] - E_D[y])^2$  is simply a consequence of the square in squared loss.

We now show that the same decomposition applies to zero-one loss in two-class problems, with  $c_1$  reflecting the fact that on noisy examples the non-optimal prediction is the correct one, and  $c_2$  reflecting that variance increases error on biased examples but decreases it on unbiased ones. Let  $P_D(y = y_*)$  be the probability over training sets in  $D$  that the learner predicts the optimal class for  $x$ .

**Theorem 2** *Equation 1 is valid for zero-one loss in two-class problems, with  $c_1 = 2P_D(y = y_*) - 1$  and  $c_2 = 1$  if  $y_m = y_*$ ,  $c_2 = -1$  otherwise.*

*Proof.*  $L(a, b)$  represents zero-one loss throughout this proof. We begin by showing that

$$E_t[L(t, y)] = L(y_*, y) + c_0 E_t[L(t, y_*)] \quad (3)$$

with  $c_0 = 1$  if  $y = y_*$  and  $c_0 = -1$  if  $y \neq y_*$ . If  $y = y_*$  Equation 3 is trivially true with  $c_0 = 1$ . Assume now that  $y \neq y_*$ . Given that there are only two classes, if  $y \neq y_*$  then  $t \neq y_*$  implies that  $t = y$  and vice-versa. Therefore  $P_t(t = y) = P_t(t \neq y_*)$ , and

$$\begin{aligned} E_t[L(t, y)] &= P_t(t \neq y) = 1 - P_t(t = y) = 1 - P_t(t \neq y_*) = 1 - E_t[L(t, y_*)] \\ &= L(y_*, y) - E_t[L(t, y_*)] = L(y_*, y) + c_0 E_t[L(t, y_*)] \end{aligned} \quad (4)$$

with  $c_0 = -1$ , proving Equation 3. We now show in a similar manner that

$$E_D[L(y_*, y)] = L(y_*, y_m) + c_2 E_D[L(y_m, y)] \quad (5)$$

with  $c_2 = 1$  if  $y_m = y_*$  and  $c_2 = -1$  if  $y_m \neq y_*$ . If  $y_m = y_*$  Equation 5 is trivially true with  $c_2 = 1$ . If  $y_m \neq y_*$  then  $y_m \neq y$  implies that  $y_* = y$  and vice-versa, and

$$\begin{aligned} E_D[L(y_*, y)] &= P_D(y_* \neq y) = 1 - P_D(y_* = y) = 1 - P_D(y_m \neq y) = 1 - E_D[L(y_m, y)] \\ &= L(y_*, y_m) - E_D[L(y_m, y)] = L(y_*, y_m) + c_2 E_D[L(y_m, y)] \end{aligned} \quad (6)$$

with  $c_2 = -1$ , proving Equation 5. Using Equation 3,

$$E_{D,t}[L(t, y)] = E_D[E_t[L(t, y)]] = E_D[L(y_*, y) + c_0 E_t[L(t, y_*)]] \quad (7)$$

Since  $L(t, y_*)$  does not depend on  $D$ ,

$$E_{D,t}[L(t, y)] = E_D[c_0]E_t[L(t, y_*)] + E_D[L(y_*, y)] \quad (8)$$

and since

$$E_D[c_0] = P_D(y = y_*) - P_D(y \neq y_*) = 2P_D(y = y_*) - 1 = c_1 \quad (9)$$

we finally obtain Equation 1, using Equation 5.  $\square$

This decomposition for zero-one loss is closely related to that of Kong and Dietterich (1995). The main differences are that Kong and Dietterich ignored the noise component  $N(x)$  and defined variance simply as the difference between loss and bias, apparently unaware that the absolute value of that difference is the average loss incurred relative to the most frequent prediction. A side-effect of this is that Kong and Dietterich incorporate  $c_2$  into their definition of variance, which can therefore be negative. Kohavi and Wolpert (1996) and others have criticized this fact, since variance for squared loss must be positive. However, our decomposition shows that the subtractive effect of variance follows from a self-consistent definition of bias and variance for zero-one and squared loss, even if the variance itself remains positive. The fact that variance is additive in unbiased examples but subtractive in biased ones has significant consequences. If a learner is biased on an example, increasing variance decreases loss. This behavior is markedly different from that of squared loss, but is obtained with the same definitions of bias and variance, purely as a result of the different properties of zero-one loss. It helps explain how highly unstable learners like decision-tree and rule induction algorithms can produce excellent results in practice, even given very limited quantities of data. In effect, when zero-one loss is the evaluation criterion, there is a much higher tolerance for variance than if the bias-variance decomposition was purely additive, because the increase in average loss caused by variance on unbiased examples is partly offset (or more than offset) by its decrease on biased ones. The average loss over all examples is the sum of noise, the average bias and what might be termed the *net variance*,  $E_x[(1 - 2B(x))V(x)]$ :

$$E_{D,t,x}[L(t, y)] = E_x[c_1 N(x)] + E_x[B(x)] + E_x[(1 - 2B(x))V(x)] \quad (10)$$

by averaging Equation 1 over all test examples  $x$ .

The  $c_1$  factor (see Equation 9) also points to a key difference between zero-one and squared loss. In squared loss, increasing noise always increases error. In zero-one loss, for training sets and test examples where  $y \neq y_*$ , increasing noise decreases error, and a high noise level can therefore in principle be beneficial to performance.

The same decomposition applies in the more general case of multiclass problems, with correspondingly generalized coefficients  $c_1$  and  $c_2$ .

**Theorem 3** *Equation 1 is valid for zero-one loss in multiclass problems, with  $c_1 = P_D(y = y_*) - P_D(y \neq y_*) P_t(y = t \mid y_* \neq t)$  and  $c_2 = 1$  if  $y_m = y_*$ ,  $c_2 = -P_D(y = y_* \mid y \neq y_m)$  otherwise.*

*Proof.* The proof is similar to that of Theorem 2, with the key difference that now  $y \neq y_*$  and  $t \neq y_*$  no longer imply that  $t = y$ , and  $y_m \neq y_*$  and  $y_m \neq y$  no longer imply that  $y = y_*$ . Given that  $y \neq y_*$  implies  $P_t(y = t \mid y_* = t) = 0$ ,

$$\begin{aligned} P_t(y = t) &= P_t(y_* \neq t) P_t(y = t \mid y_* \neq t) + P_t(y_* = t) P_t(y = t \mid y_* = t) \\ &= P_t(y_* \neq t) P_t(y = t \mid y_* \neq t) \end{aligned} \quad (11)$$

and Equation 4 becomes

$$\begin{aligned} E_t[L(t, y)] &= P_t(y \neq t) = 1 - P_t(y = t) = 1 - P_t(y_* \neq t) P_t(y = t \mid y_* \neq t) \\ &= L(y_*, y) + c_0 E_t[L(t, y_*)] \end{aligned} \quad (12)$$

with  $c_0 = -P_t(y = t \mid y_* \neq t)$ . When  $y = y_*$  Equation 3 is trivially true with  $c_0 = 1$ , as before. A similar treatment applies to Equation 5, leading to  $c_2 = -P_D(y = y_* \mid y \neq y_m)$  if  $y_m \neq y_*$ , etc. Given that

$$E_D[c_0] = P_D(y = y_*) - P_D(y \neq y_*) P_t(y = t \mid y_* \neq t) = c_1 \quad (13)$$

we obtain Theorem 3.  $\square$

Theorem 3 means that in multiclass problems not all variance on biased examples contributes to reducing loss; of all training sets for which  $y \neq y_m$ , only some have  $y = y_*$ , and it is in these that loss is reduced. This leads to an interesting insight: when zero-one loss is the evaluation criterion, the tolerance for variance will decrease as the number of classes increases, other things being equal. Thus the ideal setting for the “bias-variance trade-off” parameter in a learner (e.g., the number of neighbors in  $k$ -nearest neighbor) may be more in the direction of high variance in problems with fewer classes.

In many classification problems, zero-one loss is an inappropriate evaluation measure because misclassification costs are asymmetric; for example, classifying a cancerous patient as healthy is likely to be more costly than the reverse. Consider the two class case with



$\forall_y L(y, y) = 0$  (i.e., there is no cost for making the correct prediction), and with any nonzero real values for  $L(y_1, y_2)$  when  $y_1 \neq y_2$ . The decomposition above also applies in this case, with the appropriate choice of  $c_1$  and  $c_2$ .

**Theorem 4** *In two-class problems, Equation 1 is valid for any real-valued loss function for which  $\forall_y L(y, y) = 0$  and  $\forall_{y_1 \neq y_2} L(y_1, y_2) \neq 0$ , with  $c_1 = P_D(y = y_*) - \frac{L(y_*, y)}{L(y, y_*)} P_D(y \neq y_*)$  and  $c_2 = 1$  if  $y_m = y_*$ ,  $c_2 = -\frac{L(y_*, y_m)}{L(y_m, y_*)}$  otherwise.*

*Proof.* We begin by showing that

$$L(t, y) = L(y_*, y) + c_0 L(t, y_*) \quad (14)$$

with  $c_0 = 1$  if  $y = y_*$  and  $c_0 = -\frac{L(y_*, y)}{L(y, y_*)}$  otherwise. If  $y = y_*$  Equation 14 is trivially true with  $c_0 = 1$ . If  $t = y_*$ ,  $L(t, y) = L(y_*, y) - \frac{L(y_*, y)}{L(y, y_*)} L(t, y_*)$  is true because it reduces to  $L(t, y) = L(t, y) - 0$ . If  $t = y$ ,  $L(t, y) = L(y_*, y) - \frac{L(y_*, y)}{L(y, y_*)} L(t, y_*)$  is true because it reduces to  $L(t, t) = L(y_*, y) - L(y_*, y)$ , or  $0 = 0$ . But if  $y \neq y_*$  and we have a two-class problem, either  $t = y_*$  or  $t = y$  must be true. Therefore if  $y \neq y_*$  it is always true that  $L(t, y) = L(y_*, y) - \frac{L(y_*, y)}{L(y, y_*)} L(t, y_*)$ , completing the proof of Equation 14. We now show in a similar manner that

$$L(y_*, y) = L(y_*, y_m) + c_2 L(y_m, y) \quad (15)$$

with  $c_2 = 1$  if  $y_m = y_*$  and  $c_2 = -\frac{L(y_*, y_m)}{L(y_m, y_*)}$  otherwise. If  $y_m = y_*$  Equation 15 is trivially true with  $c_2 = 1$ . If  $y = y_m$ ,  $L(y_*, y) = L(y_*, y_m) - \frac{L(y_*, y_m)}{L(y_m, y_*)} L(y_m, y)$  is true because it reduces to  $L(y_*, y_m) = L(y_*, y_m) - 0$ . If  $y = y_*$ ,  $L(y_*, y) = L(y_*, y_m) - \frac{L(y_*, y_m)}{L(y_m, y_*)} L(y_m, y)$  is true because it reduces to  $L(y_*, y_*) = L(y_*, y_m) - L(y_*, y_m)$ , or  $0 = 0$ . But if  $y_m \neq y_*$  and we have a two-class problem, either  $y = y_m$  or  $y = y_*$  must be true. Therefore if  $y_m \neq y_*$  it is always true that  $L(y_*, y) = L(y_*, y_m) - \frac{L(y_*, y_m)}{L(y_m, y_*)} L(y_m, y)$ , completing the proof of Equation 15. Using Equation 14, and considering that  $L(y_*, y)$  and  $c_0$  do not depend on  $t$  and  $L(t, y_*)$  does not depend on  $D$ ,

$$\begin{aligned} E_{D,t}[L(t, y)] &= E_D[E_t[L(t, y)]] = E_D[L(y_*, y) + c_0 E_t[L(t, y_*)]] \\ &= E_D[L(y_*, y)] + E_D[c_0] E_t[L(t, y_*)] \end{aligned} \quad (16)$$

Substituting Equation 15 and considering that  $E_D[c_0] = P_D(y = y_*) - \frac{L(y_*, y)}{L(y, y_*)} P_D(y \neq y_*) = c_1$  results in Equation 1.  $\square$

Theorem 4 essentially shows that the loss-reducing effect of variance on biased examples will be greater or smaller depending on how asymmetric the costs are, and on which direction

they are greater in. Whether this decomposition applies in the multiclass case is an open problem. It does not apply if  $L(y, y) \neq 0$ ; in this case the decomposition contains an additional term corresponding to the cost of the correct predictions.

### 3 Properties of the Unified Decomposition

One of the main concepts Breiman (1996a) used to explain why the bagging ensemble method reduces zero-one loss was that of an *order-correct* learner.

**Definition 5** (*Breiman, 1996a*) *A learner is order-correct on an example  $x$  iff  $\forall_{y \neq y_*} P_D(y) < P_D(y_*)$ .*

Breiman showed that bagging transforms an order-correct learner into a nearly optimal one. An order-correct learner is an unbiased one according to Definition 2:

**Theorem 5** *A learner is order-correct on an example  $x$  iff  $B(x) = 0$  under zero-one loss.*

The proof is immediate from the definitions, considering that  $y_m$  for zero-one loss is the most frequent prediction.

Schapire et al. (1997) proposed an explanation for why the boosting ensemble method works in terms of the notion of *margin*. For algorithms like bagging and boosting, that generate multiple hypotheses by applying the same learner to multiple training sets, their definition of margin can be stated as follows.

**Definition 6** (*Schapire et al., 1997*) *In two-class problems, the margin of a learner on an example  $x$  is  $M(x) = P_D(y = t) - P_D(y \neq t)$ .*

A positive margin indicates a correct classification by the ensemble, and a negative one an error. Intuitively, a large margin corresponds to a high confidence in the prediction.  $D$  here is the set of training sets to which the learner is applied. For example, if 100 rounds of boosting are carried out,  $|D| = 100$ . Further, for algorithms like boosting where the different training sets (and corresponding predictions) have different weights that sum to 1,  $P_D(\cdot)$  is computed according to these weights. Definitions 1–4 apply unchanged in this situation. In effect, we have generalized the notions of bias and variance to apply to any training set selection scheme, not simply the traditional one of “all possible training sets of a given size, with equal weights.”

Schapire et al. (1997) showed that it is possible to bound an ensemble’s generalization error (i.e., its zero-one loss on test examples) in terms of the distribution of margins on training examples and the VC dimension of the base learner. In particular, the smaller the probability of a low margin, the lower the bound on generalization error. The following theorem shows that the margin is closely related to bias and variance as defined above.

**Theorem 6** *The margin of a learner on an example  $x$  can be expressed in terms of its zero-one bias and variance as  $M(x) = \pm[2B(x) - 1][2V(x) - 1]$ , with positive sign if  $y_* = t$  and negative sign otherwise.*

*Proof.* When  $y_* = t$ ,  $M(x) = P_D(y = y_*) - P_D(y \neq y_*) = 2P_D(y = y_*) - 1$ . If  $B(x) = 0$ ,  $y_m = y_*$  and  $M(x) = 2P_D(y = y_m) - 1 = 2[1 - V(x)] - 1 = -[2V(x) - 1]$ . If  $B(x) = 1$  then  $M(x) = 2V(x) - 1$ . Therefore  $M(x) = [2B(x) - 1][2V(x) - 1]$ . The demonstration for  $y_* \neq t$  is similar, with  $M(x) = P_D(y \neq y_*) - P_D(y = y_*)$ .  $\square$

Conversely, it is possible to express the bias and variance in terms of the margin:  $B(x) = \frac{1}{2}[1 \pm \text{sign}(M(x))]$ ,  $V(x) = \frac{1}{2}[1 \pm |M(x)|]$ , with positive sign if  $y_* \neq t$  and negative sign otherwise. The relationship between margins and bias/variance expressed in Theorem 6 implies that Schapire et al.’s theorems can be stated in terms of the bias and variance on training examples. Bias-variance decompositions relate a learner’s loss on an example to its bias and variance on that example. However, to our knowledge this is the first time that *generalization* error is related to bias and variance on *training* examples.

Theorem 6 also sheds light on the polemic between Breiman (1996b, 1997) and Schapire et al. (1997) on how the success of ensemble methods like bagging and boosting is best explained. Breiman has argued for a bias-variance explanation, while Schapire et al. have argued for a margin-based explanation. Theorem 6 shows that these are two faces of the same coin, and helps to explain why the bias-variance explanation sometimes seems to fail when applied to boosting. Maximizing margins is a combination of reducing the number of biased examples, decreasing the variance on unbiased examples, and increasing it on biased ones (for examples where  $y_* = t$ ; the reverse, otherwise). Without differentiating between these effects it is hard to understand how boosting affects bias and variance.

Unfortunately, there are many loss functions to which the decomposition in Equation 1 does not apply. For example, it is easily shown that it does not apply to  $L(t, y) = (t - y)^m$  with arbitrary  $m$ ; in particular, it does not apply to absolute loss. However, as long as the loss function is a metric, it can be bounded from above and below by simple functions of the bias, variance and noise.

**Theorem 7** *The following inequalities are valid for any metric loss function:*

$$\begin{aligned} E_{D,t}[L(t, y)] &\leq N(x) + B(x) + V(x) \\ E_{D,t}[L(t, y)] &\geq \max(\{N(x) - B(x) - V(x), B(x) - V(x) - N(x), V(x) - B(x) - N(x)\}) \end{aligned}$$

*Proof.* Recall that a function of two arguments  $d(a_1, a_2)$  is a metric iff  $\forall_{a,b} d(a, b) \geq d(a, a) = 0$  (minimality),  $\forall_{a,b} d(a, b) = d(b, a)$  (symmetry), and  $\forall_{a,b,c} d(a, b) + d(b, c) \geq d(a, c)$  (triangle inequality). Using the triangle inequality,

$$L(t, y) \leq L(t, y_*) + L(y_*, y) \leq L(t, y_*) + L(y_*, y_m) + L(y_m, y) \quad (17)$$

Taking the expected value of this equation with respect to  $D$  and  $t$  and simplifying produces the upper bound. Using the triangle inequality and symmetry,

$$L(y_*, y_m) \leq L(y_*, t) + L(t, y) + L(y, y_m) \leq L(t, y_*) + L(t, y) + L(y_m, y) \quad (18)$$

Rearranging terms, taking the expectation wrt  $D$  and  $t$  and simplifying leads to  $E_{D,t}[L(t, y)] \geq B(x) - V(x) - N(x)$ . The remaining components of the lower bound are obtained in a similar manner.  $\square$

## 4 Experiments

We applied the bias-variance decomposition of zero-one loss proposed here in a series of experiments with classification algorithms. To our knowledge this is the most extensive such study to date, in terms of the number of datasets and number of algorithms/parameter settings studied. This section summarizes the results. We used the following 30 datasets from the UCI repository (Blake & Merz, 2000): annealing, audiology, breast cancer (Ljubljana), chess (king-rook vs. king-pawn), credit (Australian), diabetes, echocardiogram, glass, heart disease (Cleveland), hepatitis, horse colic, hypothyroid, iris, labor, LED, lenses, liver disorders, lung cancer, lymphography, mushroom, post-operative, primary tumor, promoters, solar flare, sonar, soybean (small), splice junctions, voting records, wine, and zoology.

As the noise level  $N(x)$  is very difficult to estimate, we followed previous authors (e.g., Kohavi & Wolpert (1996)) in assuming  $N(x) = 0$ . This is not too detrimental to the significance of the results because we are mainly interested in the variation of bias and variance with several factors, not their absolute values. We estimated bias, variance and zero-one loss by the following method. We randomly divided each dataset into training data

(two thirds of the examples) and test data (one third). For each dataset, we generated 100 different training sets by the *bootstrap* method (Efron & Tibshirani, 1993): if the training data consists of  $n$  examples, we create a *bootstrap replicate* of it by taking  $n$  samples *with replacement* from it, with each example having a probability of  $1/n$  of being selected at each turn. As a result, some of the examples will appear more than once in the training set, and some not at all. The 100 training sets thus obtained were taken as a sample of the set  $D$ , with  $D$  being the set of all training sets of size  $n$ . A model was then learned on each training set. We used the predictions made by these models on the test examples to estimate average zero-one loss, average bias and net variance, as defined in Section 2. We also measured the total contribution to average variance from unbiased examples  $V_u = \frac{1}{n}[\sum_{i=1}^n (1 - B(x_i))V(x_i)]$  and the contribution from biased examples  $V_b = \frac{1}{n}[\sum_{i=1}^n cB(x_i)V(x_i)]$ , where  $x_i$  is a test example,  $n$  is the number of test examples,  $c = 1$  for two-class problems, and  $c = P_D(y = y_* | y \neq y_m)$  for multiclass problems (see Theorem 3), estimated from the test set. The net variance is the difference of the two:  $V = V_u - V_b$ .

We carried out experiments with decision-tree induction, boosting, and  $k$ -nearest neighbor; their results are reported in turn. In the interests of space, we do not present the full results; instead, we summarize the main observations, and present representative examples. The complete set of experimental results obtained is available as an online appendix at <http://www.cs.washington.edu/homes/pedrod/bvd.html>.

## 4.1 Decision-Tree Induction

We used the C4.5 decision tree learner, release 8 (Quinlan, 1993). We measured zero-one loss, bias and variance while varying C4.5’s pruning parameter (the confidence level CF) from 0% (maximum pruning) to 100% (minimum) in 5% steps. The default setting is 25%. Surprisingly, we found that in most datasets CF has only a minor effect on bias and variance (and therefore loss). Only at the CF=0% extreme, where the tree is pruned all the way to the root, is there a major impact, with very high bias and loss; but this disappears by CF=5%. Figure 1 shows a typical example. These results suggest there may be room for improvement in C4.5’s pruning method (cf. Oates & Jensen (1997)).

In order to obtain a clearer picture of the bias-variance trade-off in decision tree induction, we replaced C4.5’s native pruning scheme with a limit on the number of levels allowed in the tree. (When a maximum level of  $m$  is set, every path in the tree of length greater than  $m$  is pruned back to a length of  $m$ .) The dominant effect observed is the rapid decrease of bias in the first few levels, after which it typically stabilizes. In 9 of the 25 datasets where this

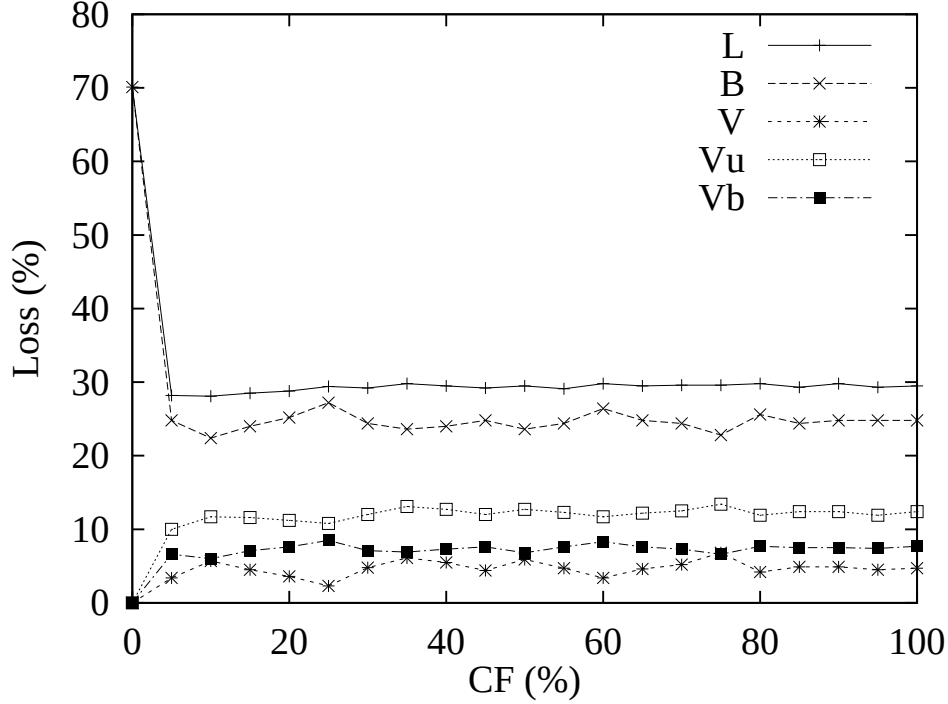


Figure 1: Effect of varying C4.5's pruning parameter: diabetes.

occurs, bias in fact increases after this point (slightly in 6, markedly in 3). In 5 datasets bias increases with the number of levels overall; in 3 of these (echocardiogram, post-operative and sonar) it increases markedly. Variance increases with the number of levels in 26 datasets; in 17 of these the increase is generally even, and much slower than the initial decrease in bias. Less-regular patterns occur in the remaining 9 datasets.  $V_u$  and  $V_b$  tend to be initially similar, but  $V_b$  increases more slowly than  $V_u$ , or decreases. At any given level,  $V_b$  typically offsets a large fraction of  $V_u$ , making variance a smaller contributor to loss than would be the case if its effect was always positive. This leads to the hypothesis that higher-variance algorithms (or settings) may be better suited to classification (zero-one loss) than regression (squared loss). Perhaps not coincidentally, research in classification has tended to explore higher-variance algorithms than research in regression.

Representative examples of the patterns observed are shown in Figure 2, where the highest level shown is the highest produced by C4.5 when it runs without any limits. Overall, the expected pattern of a trade-off in bias and variance leading to a minimum of loss at an intermediate level was observed in only 10 datasets; in 6 a decision stump was best, and in 14 an unlimited number of levels was best.

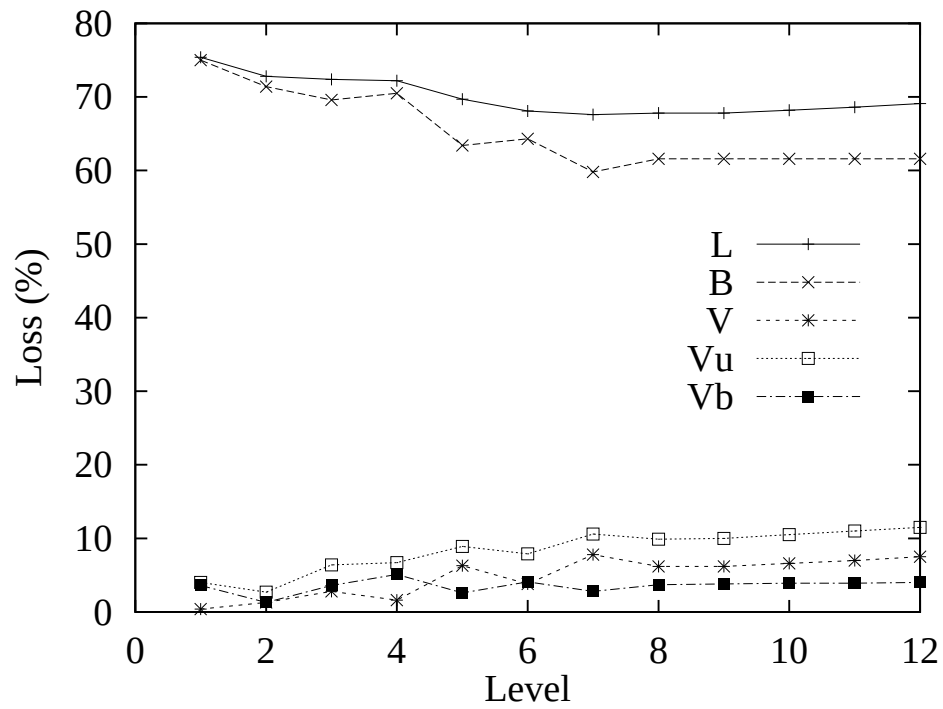
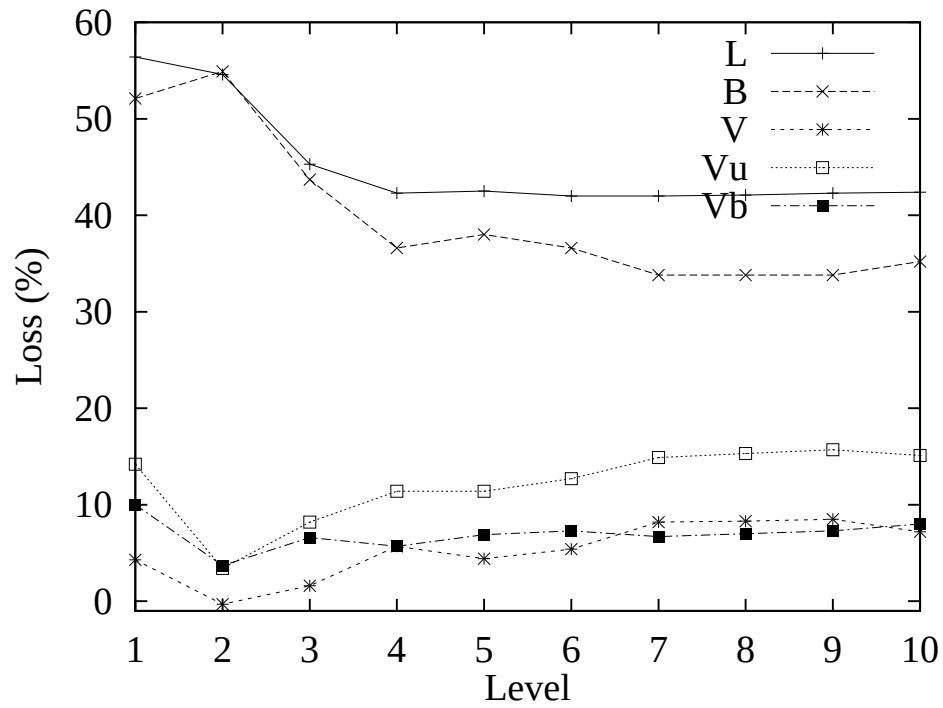


Figure 2: Effect of varying the number of levels in C4.5 trees: glass (top) and primary tumor (bottom).

## 4.2 Boosting

We also experimented with applying AdaBoost (Freund & Schapire, 1996) to C4.5. We allowed a maximum of 100 rounds of boosting. (In most datasets, loss and its components stabilized by the 20th round, and only this part is graphed.) Boosting decreases loss in 21 datasets and increases it in 3; it has no effect in the remainder. It decreases bias in 14 datasets and increases it in 7, while decreasing net variance in 18 datasets and increasing it in 7. The bulk of bias reduction typically occurs in the first few rounds. Variance reduction tends to be more gradual. On average (over all datasets) variance reduction is a much larger contributor to loss reduction than bias reduction (2.5% vs. 0.6%). Over all datasets, the variance reduction is significant at the 5% level according to sign and Wilcoxon tests, but bias reduction is not. Thus variance reduction is clearly the dominant effect when boosting is applied to C4.5; this is consistent with the notion that C4.5 is a “strong” learner. Boosting tends to reduce both  $V_u$  and  $V_b$ , but it reduces  $V_u$  much more strongly than  $V_b$  (3.0% vs. 0.5%). The ideal behavior would be to reduce  $V_u$  and increase  $V_b$ ; it may be possible to design a variant of boosting that achieves this, and as a result further reduces loss. Examples of the boosting behaviors observed are shown in Figure 3.

## 4.3 $K$ -Nearest Neighbor

We studied the bias and variance of the  $k$ -nearest neighbor algorithm (Cover & Hart, 1967) as a function of  $k$ , the number of neighbors used to predict a test example’s class. We used Euclidean distance for numeric attributes and overlap for symbolic ones.  $k$  was varied from 1 to 21 in increments of 2; typically only small values of  $k$  are used, but this extended range allows a clearer observation of its effect. The pattern of an increase in bias and a decrease in variance with  $k$  producing a minimal loss at an intermediate value of  $k$  is seldom observed; more often one of the two effects dominates throughout. In several cases bias and variance vary in the same direction with  $k$ . In 13 datasets, the lowest loss is obtained with  $k = 1$ , and in 11 with the maximum  $k$ . On average (over all datasets) bias increases markedly with  $k$  (by 4.9% from  $k = 1$  to  $k = 21$ ), but variance decreases only slightly (0.8%), resulting in much increased loss. This contradicts Friedman’s (1997) hypothesis (based on approximate analysis and artificial data) that very large values of  $k$  should be beneficial. This may be attributable to the fact that, as  $k$  increases, what Friedman calls the “boundary bias” changes from negative to positive for a majority of the examples, wiping out the benefits of low variance. Interestingly, increasing  $k$  in  $k$ -NN has the “ideal” effect of reducing  $V_u$  (by



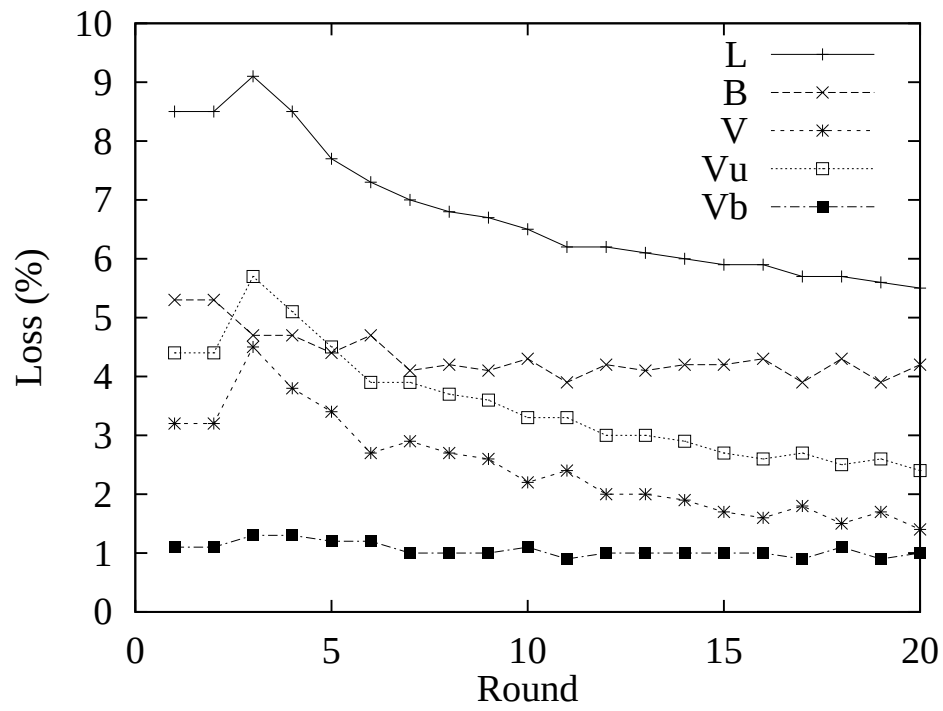
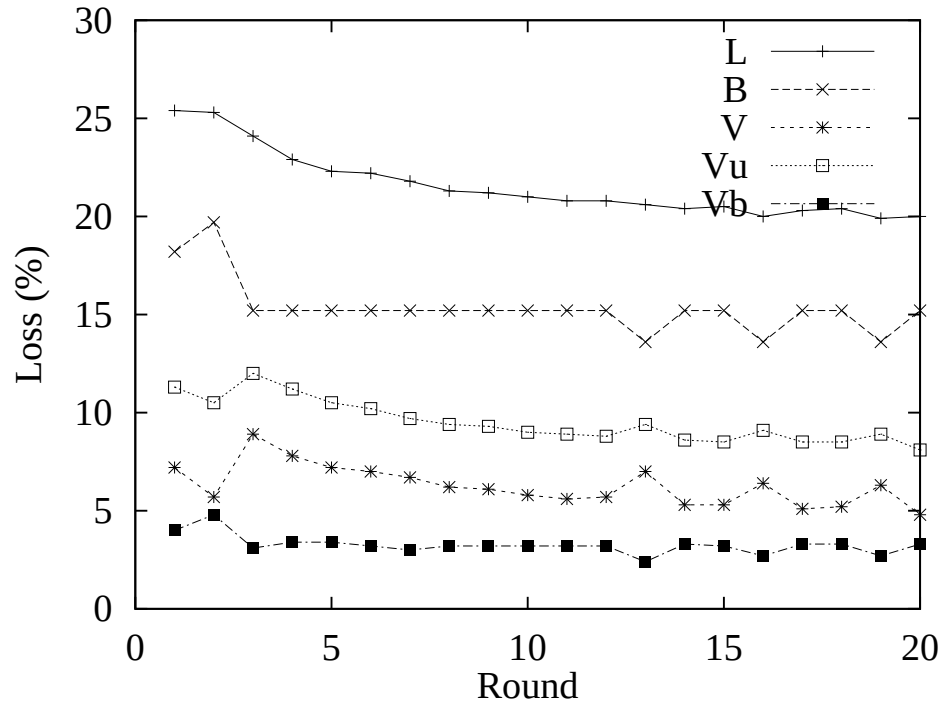


Figure 3: Effect of boosting on C4.5: audiology (top) and splice junctions (bottom).

0.5% on average) while increasing  $V_b$  (0.3%). Figure 4 shows examples of the different types of behavior observed.

When  $k$ -nearest neighbor is applied to problems with many classes, variance can at first increase with  $k$  (for  $k \leq 10$ , very roughly) due to an increase in the number of randomly-resolved ties. This confirms Kohavi and Wolpert’s (1996) observations. However, when  $k$  is increased further variance starts to decrease, as it must in the long run, given that the variance for  $k$  equal to the training set size should be close to zero (except when two or more class priors are very close). See the graph for the audiology dataset in Figure 4.

## 5 Related Work

The first bias-variance decomposition for zero-one loss was proposed by Kong and Dietterich (1995). Although they proposed it in a purely *ad hoc* manner and only applied it to one ensemble learner in one artificial, noise-free domain, our results show that it is in fact a well-founded and useful decomposition, even if incomplete. Breiman (1996b) proposed a decomposition for the average zero-one loss over all examples, leaving bias and variance for a specific example  $x$  undefined. As Tibshirani (1996) points out, Breiman’s definitions of bias and variance have some undesirable properties, seeming artificially constructed to produce a purely additive decomposition. Tibshirani’s (1996) definitions do not suffer from these problems; on the other hand, he makes no use of the variance, instead decomposing zero-one loss into bias and an unrelated quantity he calls the “aggregation effect.” James and Hastie (1997) extend this approach by defining bias and variance but decomposing loss in terms of two quantities they call “systematic effect” and “variance effect.” Kohavi and Wolpert (1996) defined bias and variance in terms of quadratic functions of  $P_t(t)$  and  $P_D(y)$ . Although the resulting decomposition is purely additive, it suffers from the serious problem that it does not assign zero bias to the Bayes classifier. Also, although Kohavi and Wolpert emphasize the fact that their definition of zero-one bias is not restricted to taking on the values 0 or 1, it would seem that a binary-valued bias is the natural consequence of a binary-valued loss function. In practice, Kohavi and Wolpert’s method produces biased estimates of bias and variance; although their estimators can be debiased, this obscures their meaning (for example, the corrected bias can be negative). Friedman (1997) studied the relationship between zero-one loss and the bias and variance of class probability estimates. He emphasized that the effect of bias and variance is strongly non-additive; increasing variance can reduce error. In this article we obtain similar results directly in terms of the bias and variance of

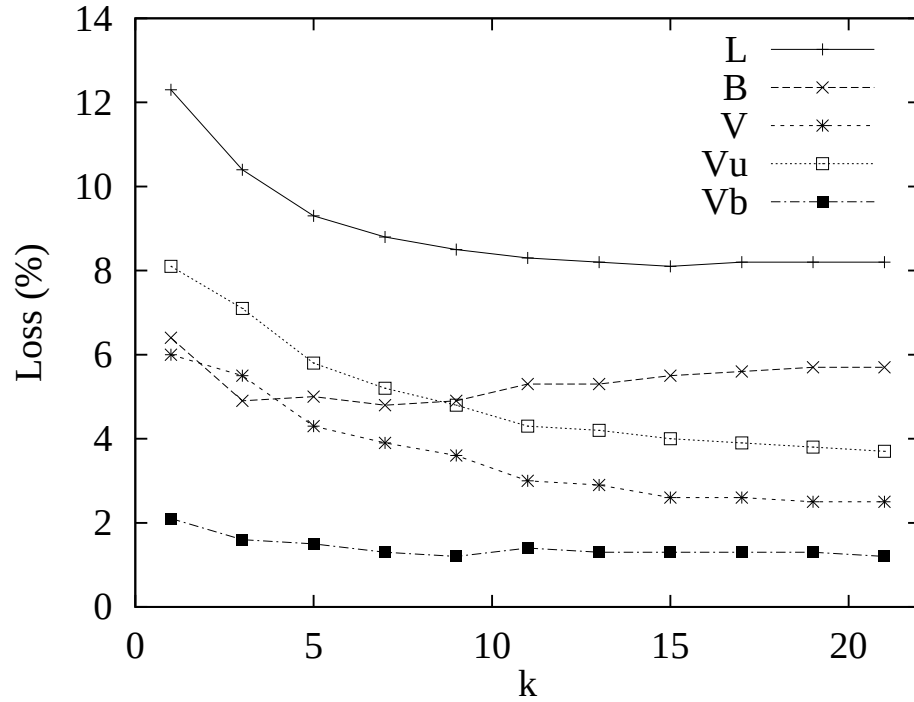
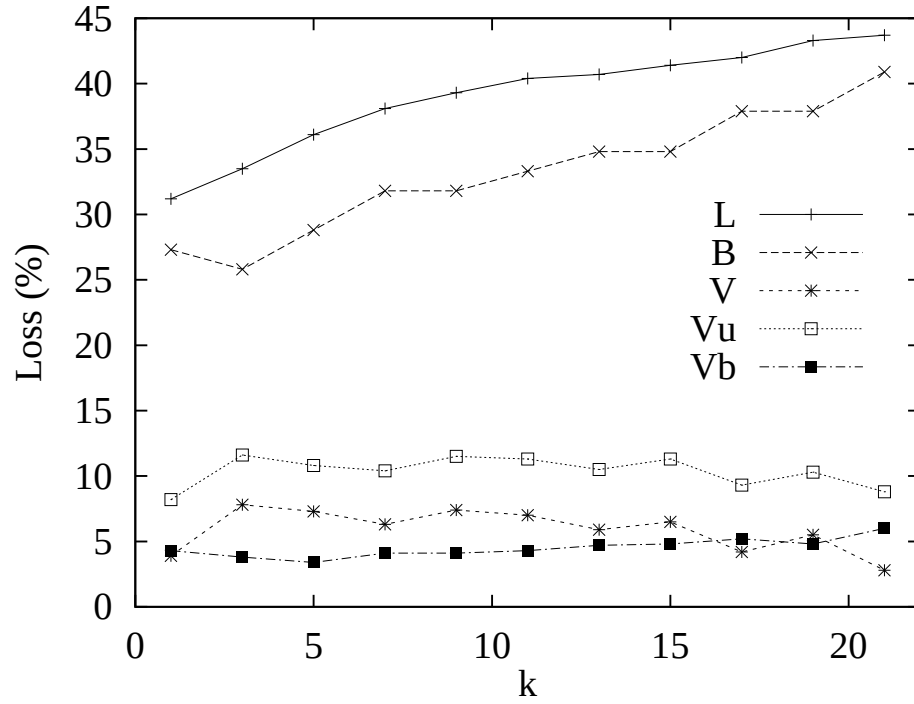


Figure 4: Effect of varying  $k$  in  $k$ -nearest neighbor: audiology (top) and chess (bottom).

class predictions, and without Friedman’s restrictive assumptions (only two classes, Gaussian probabilities).

## 6 Future Work

The main limitation of the definitions of bias and variance proposed here is that many loss functions cannot be decomposed according to them and Equation 1 (e.g., absolute loss). Since it is unlikely that meaningful definitions exist for which a simple decomposition is always possible, a central direction for future work is determining general properties of loss functions that are necessary and/or sufficient for Equation 1 to apply. Even when it does not, it may be possible to usefully relate loss to bias and variance as defined here. Another major direction for future work is applying the decomposition to a wider variety of learners, in order to gain insight about their behavior, both with respect to variations within a method and with respect to comparisons between methods. We would also like to study experimentally the effect of different domain characteristics (e.g., sparseness of the data) on the bias and variance of different learning algorithms. The resulting improved understanding should allow us to design learners that are more easily adapted to a wide range of domains.

## 7 Conclusion

In this article we proposed general definitions of bias and variance applicable to any loss function, and derived the corresponding decompositions for squared loss, zero-one loss and variable misclassification costs. While these decompositions are not always purely additive, we believe that more insight is gained from this approach—formulating consistent definitions and investigating what follows from them—than from crafting definitions case-by-case to make the decomposition purely additive. For example, uncovering the different role of variance on biased and unbiased examples in zero-one loss leads to an improved understanding of classification algorithms, and of how they differ from regression ones. We also showed that margins can be expressed as a function of zero-one bias and variance, and that a simple relationship between loss, bias and variance exists for any metric loss function. The utility of our decomposition was illustrated by an extensive empirical study of bias and variance in decision tree induction, boosting, and  $k$ -nearest neighbor learning.

C functions implementing the bias-variance decomposition proposed in this article are available at <http://www.cs.washington.edu/homes/pedrod/bvd.c>.

## Acknowledgments

This research was partly supported by a PRAXIS XXI grant. The author is grateful to all those who provided the datasets used in the experiments.

## References

- Bauer, E., & Kohavi, R. (1999). An empirical comparison of voting classification algorithms: Bagging, boosting and variants. *Machine Learning*, 36, 105–142.
- Blake, C., & Merz, C. J. (2000). *UCI repository of machine learning databases*. Department of Information and Computer Science, University of California at Irvine, Irvine, CA. <http://www.ics.uci.edu/~mlearn/MLRepository.html>.
- Breiman, L. (1996a). Bagging predictors. *Machine Learning*, 24, 123–140.
- Breiman, L. (1996b). *Bias, variance and arcing classifiers* (Technical Report 460). Statistics Department, University of California at Berkeley, Berkeley, CA.
- Breiman, L. (1997). *Arcing the edge* (Technical Report 486). Statistics Department, University of California at Berkeley, Berkeley, CA.
- Domingos, P., & Pazzani, M. (1997). On the optimality of the simple Bayesian classifier under zero-one loss. *Machine Learning*, 29, 103–130.
- Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. New York, NY: Chapman and Hall.
- Freund, Y., & Schapire, R. E. (1996). Experiments with a new boosting algorithm. *Proceedings of the Thirteenth International Conference on Machine Learning* (pp. 148–156). Bari, Italy: Morgan Kaufmann.
- Friedman, J. H. (1997). On bias, variance, 0/1 - loss, and the curse-of-dimensionality. *Data Mining and Knowledge Discovery*, 1, 55–77.
- Geman, S., Bienenstock, E., & Doursat, R. (1992). Neural networks and the bias/variance dilemma. *Neural Computation*, 4, 1–58.
- Holte, R. C. (1993). Very simple classification rules perform well on most commonly used datasets. *Machine Learning*, 11, 63–91.

- James, G., & Hastie, T. (1997). *Generalizations of the bias/variance decomposition for prediction error* (Technical Report). Department of Statistics, Stanford University, Stanford, CA.
- Kohavi, R., & Wolpert, D. H. (1996). Bias plus variance decomposition for zero-one loss functions. *Proceedings of the Thirteenth International Conference on Machine Learning* (pp. 275–283). Bari, Italy: Morgan Kaufmann.
- Kong, E. B., & Dietterich, T. G. (1995). Error-correcting output coding corrects bias and variance. *Proceedings of the Twelfth International Conference on Machine Learning* (pp. 313–321). Tahoe City, CA: Morgan Kaufmann.
- Oates, T., & Jensen, D. (1997). The effects of training set size on decision tree complexity. *Proceedings of the Fourteenth International Conference on Machine Learning* (pp. 254–262). Madison, WI: Morgan Kaufmann.
- Quinlan, J. R. (1993). *C4.5: Programs for machine learning*. San Mateo, CA: Morgan Kaufmann.
- Schapire, R. E., Freund, Y., Bartlett, P., & Lee, W. S. (1997). Boosting the margin: A new explanation for the effectiveness of voting methods. *Proceedings of the Fourteenth International Conference on Machine Learning* (pp. 322–330). Nashville, TN: Morgan Kaufmann.
- Tibshirani, R. (1996). *Bias, variance and prediction error for classification rules* (Technical Report). Department of Preventive Medicine and Biostatistics and Department of Statistics, University of Toronto, Toronto, Canada.